12. Implement an Earley parser for context-free grammars using a simple python program.

Aim

To implement a simple Earley parser in Python to check whether a given input string is accepted by a context-free grammar (CFG).

Algorithm

1. Define the context-free grammar with production rules.
2. Initialize the Earley chart with the start production.
3. Apply Predictor to add possible productions.
4. Apply Scanner to match input symbols.
5. Apply Completer to complete productions when their symbols are recognized.
6. Check whether the start production is completed at the end of the input.
7. Display whether the string is accepted or rejected.

Python Code

```
# Simple Earley Parser

def earley_parser(grammar, start, text):
    n = len(text)
    chart = [set() for _ in range(n + 1)]

    # Item: (left side, right side, dot position, origin)
    chart[0].add((start, tuple(grammar[start][0]), 0, 0))

    for i in range(n + 1):
        changed = True

        while changed:
            changed = False

            for item in list(chart[i]):
                lhs, rhs, dot, origin = item

                if dot < len(rhs):
```

```
                symbol = rhs[dot]

                # Predictor
                if symbol in grammar:
                    for rule in grammar[symbol]:
                        new_item = (symbol, tuple(rule), 0, i)
                        if new_item not in chart[i]:
                            chart[i].add(new_item)
                            changed = True

                # Scanner
                elif i < n and text[i] == symbol:
                    chart[i + 1].add(
                        (lhs, rhs, dot + 1, origin)
                    )

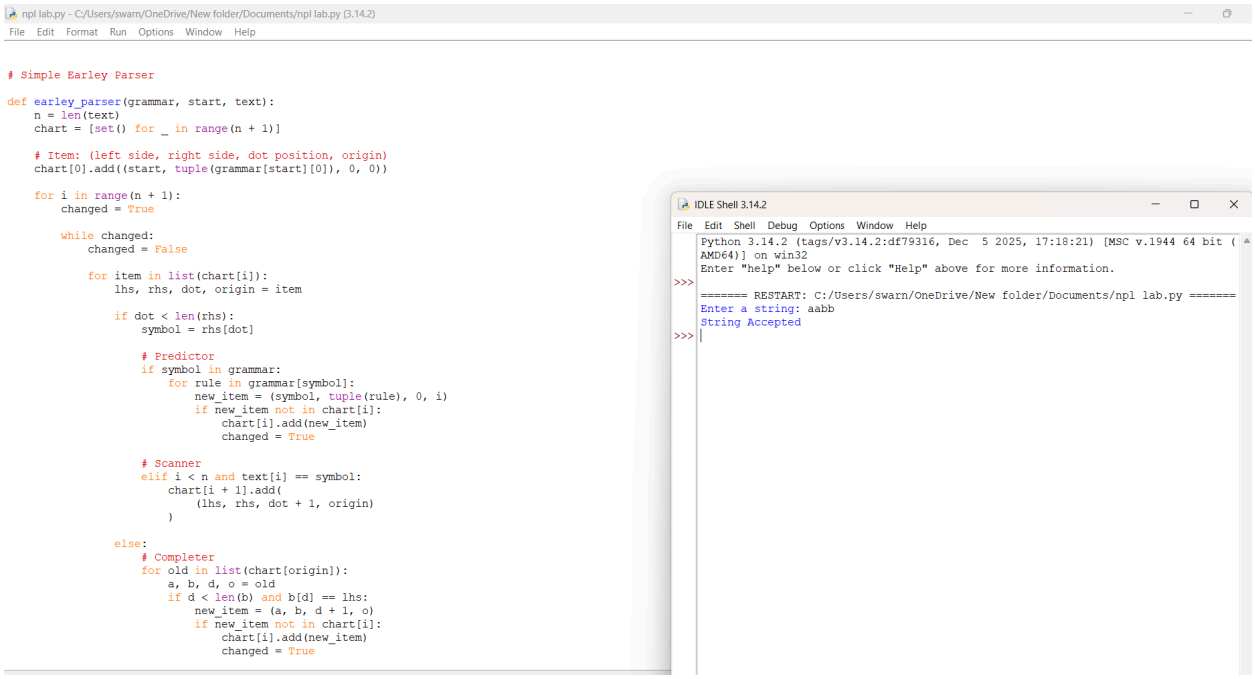
            else:
                # Completer
                for old in list(chart[origin]):
                    a, b, d, o = old
                    if d < len(b) and b[d] == lhs:
                        new_item = (a, b, d + 1, o)
                        if new_item not in chart[i]:
                            chart[i].add(new_item)
                            changed = True

    # Check completed start production
    return any(
        lhs == start and dot == len(rhs) and origin == 0
        for lhs, rhs, dot, origin in chart[n]
    )


grammar = {
    'S': [['a', 'S', 'b'], []]
}

string = input("Enter a string: ")

if earley_parser(grammar, 'S', string):
    print("String Accepted")
else:
    print("String Rejected")
```



Result

Thus, an Earley parser was implemented successfully in Python to determine whether input strings are accepted by a context-free grammar.